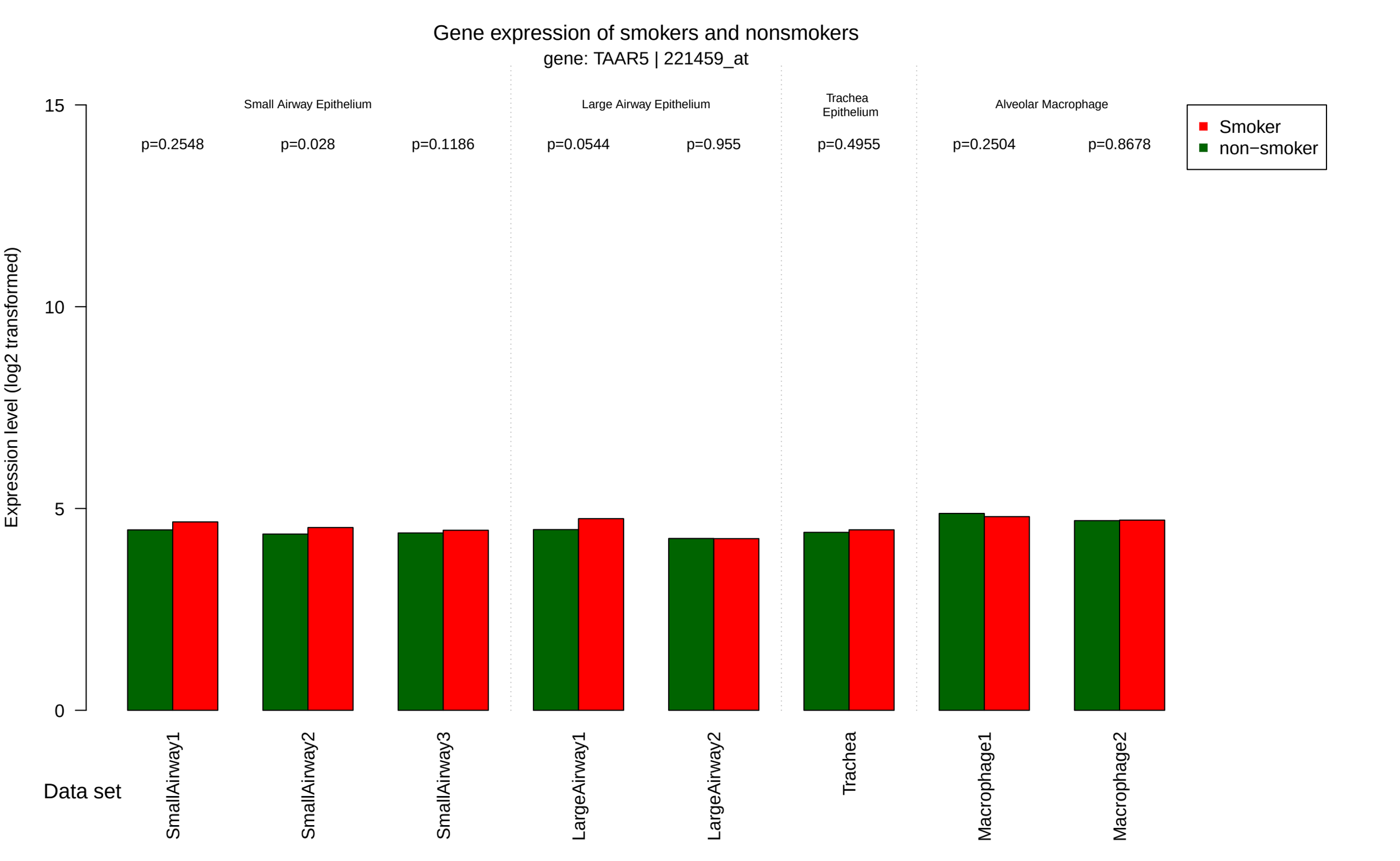
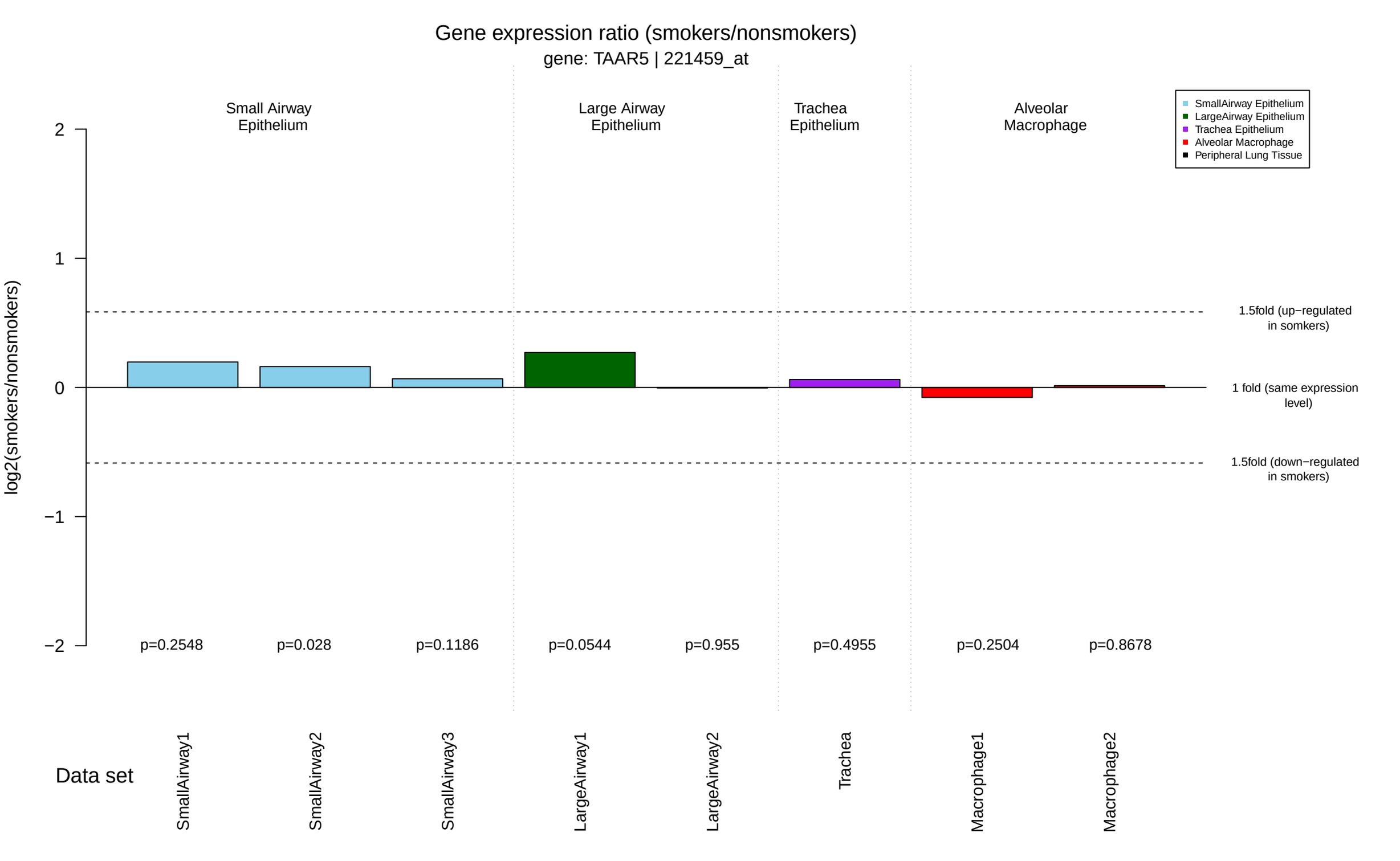
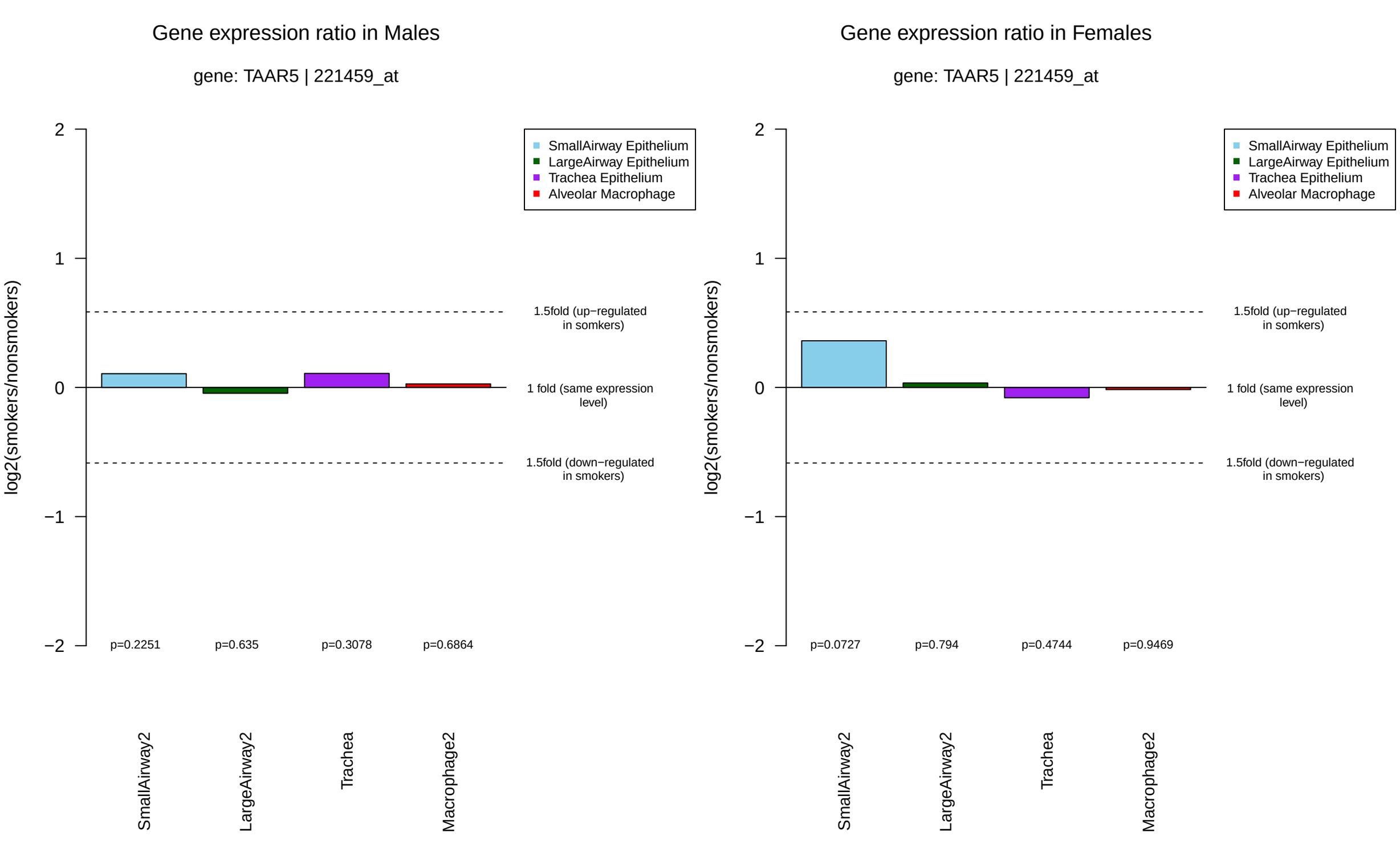
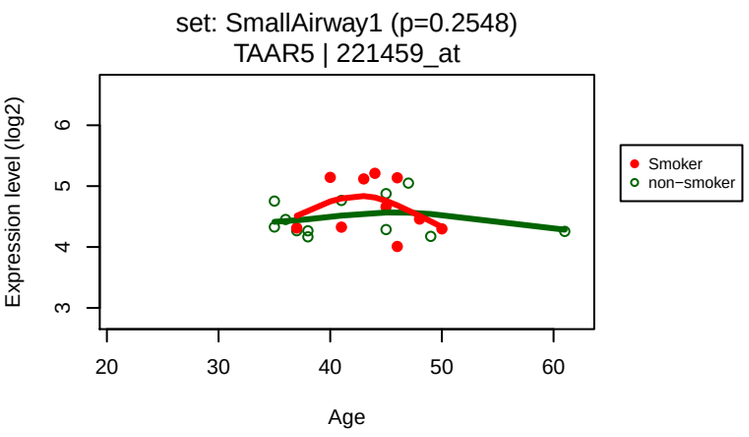
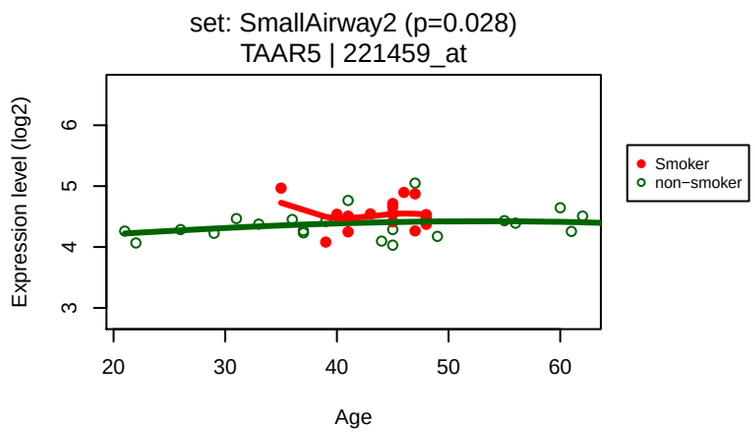
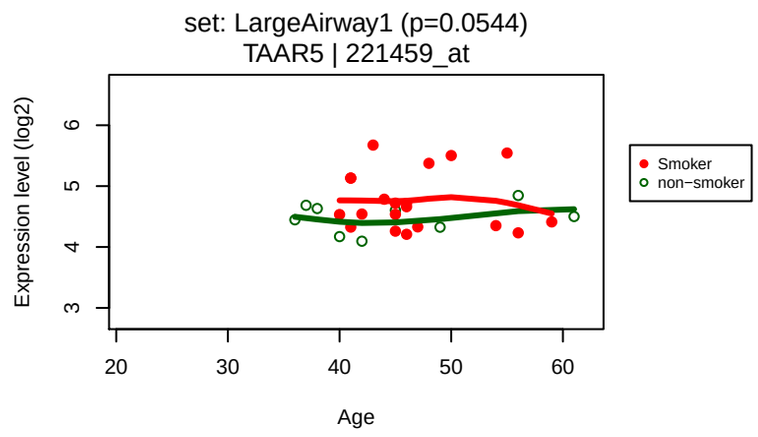

Gene expression of smokers and nonsmokers
gene: TAAR5 | 221459_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2548
p=0.028
p=0.1186
p=0.0544
p=0.955
p=0.4955
p=0.2504
p=0.8678
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

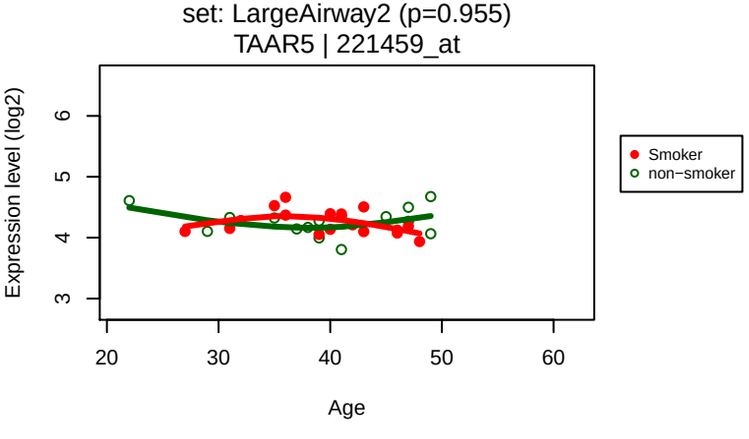

Gene expression ratio (smokers/nonsmokers)
gene: TAAR5 | 221459_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2548
p=0.028
p=0.1186
p=0.0544
p=0.955
p=0.4955
p=0.2504
p=0.8678
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

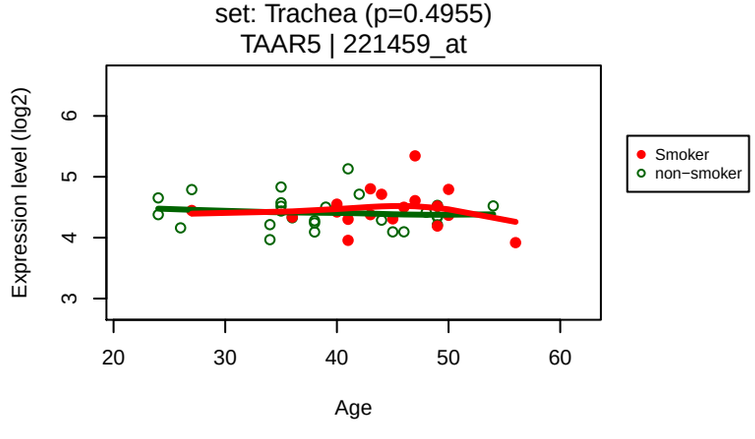

Gene expression ratio in Males
gene: TAAR5 | 221459_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2251
p=0.635
p=0.3078
p=0.6864
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TAAR5 | 221459_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0727
p=0.794
p=0.4744
p=0.9469
SmallAirway2
LargeAirway2
Trachea
Macrophage2

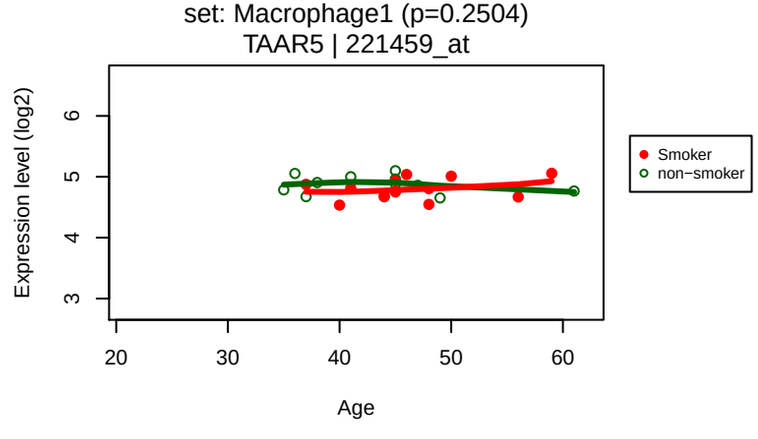

set: SmallAirway1 (p=0.2548)
TAAR5 | 221459_at
Expression level (log2)
Age
Smoker
non−smoker

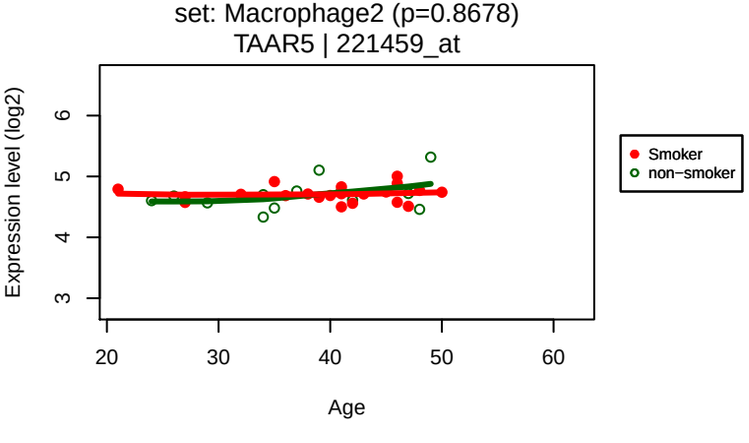

set: SmallAirway2 (p=0.028)
TAAR5 | 221459_at
Expression level (log2)
Age
Smoker
non−smoker

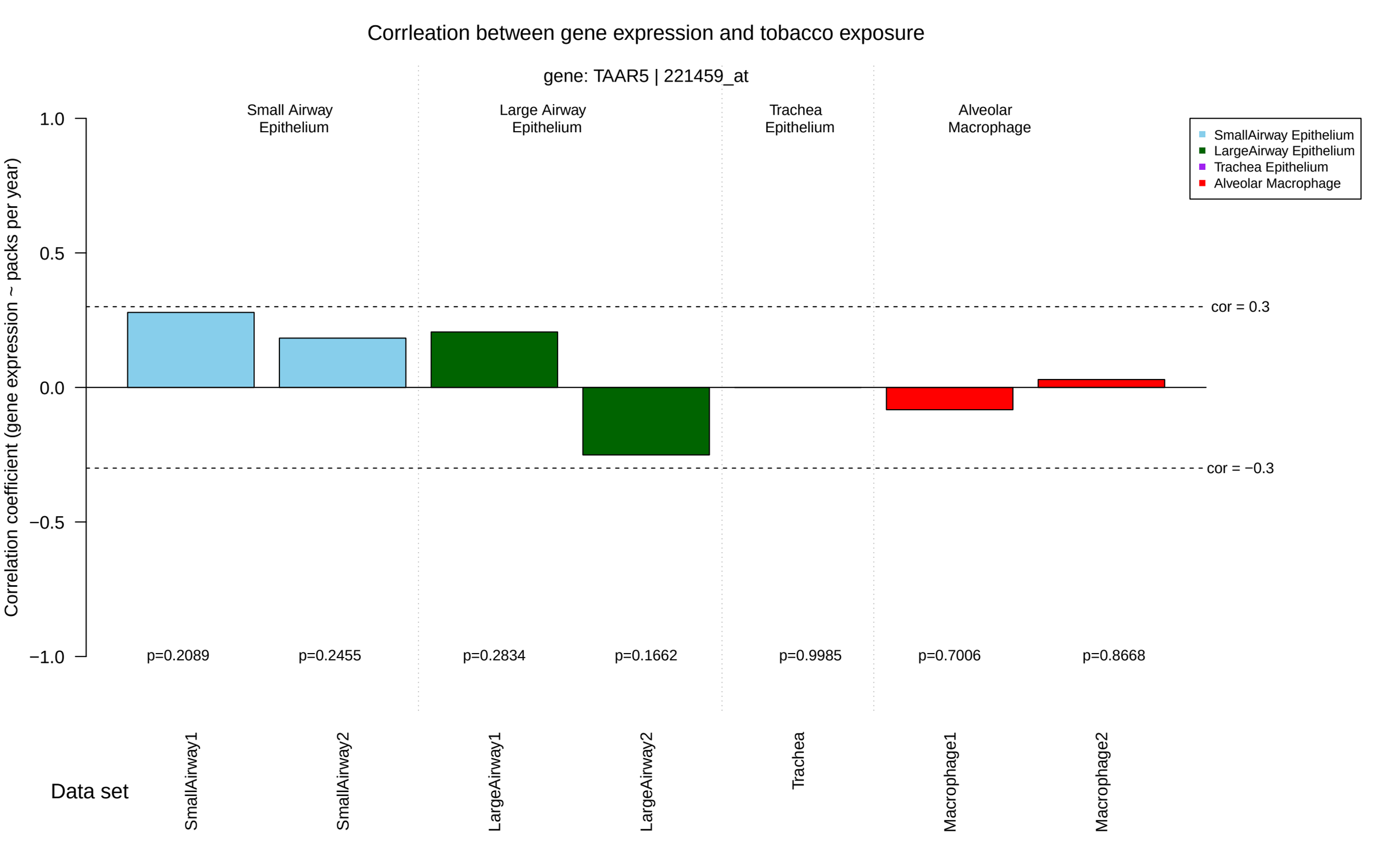

set: LargeAirway1 (p=0.0544)
TAAR5 | 221459_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.955)
TAAR5 | 221459_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4955)
TAAR5 | 221459_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2504)
TAAR5 | 221459_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8678)
TAAR5 | 221459_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TAAR5 | 221459_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2089
p=0.2455
p=0.2834
p=0.1662
p=0.9985
p=0.7006
p=0.8668
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2